<table>
<tr><td>Program:</td><td colspan="8">B. Tech. (Computer Engineering)</td><td colspan="6">Semester: VI</td></tr>
<tr><td>Course:</td><td colspan="8">Operating Systems</td><td colspan="6">Code: BCE26PC01</td></tr>
<tr><td rowspan="3">Credit</td><td colspan="4">Teaching Scheme (Hrs./Week)</td><td colspan="8">Evaluation Scheme and Marks</td></tr>
<tr><td rowspan="2">Lecture</td><td rowspan="2">Practical</td><td rowspan="2">Tutorial</td><td rowspan="2">Other</td><td colspan="2">FA</td><td rowspan="2">SA</td><td rowspan="2">TW</td><td rowspan="2">OR</td><td rowspan="2">PR</td><td rowspan="2">Total</td></tr>
<tr><td>FA1</td><td>FA2</td></tr>
<tr><td>3</td><td>2</td><td>2</td><td>-</td><td>-</td><td>10</td><td>10</td><td>30</td><td>50</td><td>-</td><td>-</td><td>100</td></tr>
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**Prior knowledge of** Computer Organization, Data Structures, basics of Linux commands, and programming **is essential**

**Course Objectives:**

This course aims at enabling students:

1. To introduce the Operating system and process, threads and CPU scheduling
2. To explore inter process communication mechanisms, to introduce the critical-section problem and solutions.
3. To explore various techniques of allocating memory to processes and explain the concepts of demand paging, page-replacement algorithms.
4. To describe the details of implementing local file systems and directory structures

**Course Outcomes:**

After learning the course, the students will be able to:

1. Compare various process scheduling algorithms for a given snapshot of the system.
2. Analyze IPC mechanisms and solutions of process synchronization for critical section problems.
3. Analyze the performance of memory management algorithms for a given problem.
4. Apply disk scheduling policies for a given I/O request sequence with file management concepts

**Laboratory related guidelines:**

**Guidelines for Students:**

1. The laboratory assignments are to be submitted by students in the form of a journal. Journal consists of Certificate, Index, and handwritten write- up of each assignment. Each assignment write - up should have Title, Objectives, Theory - Concept in brief, Algorithm/Flowchart, Test cases, Conclusion.
2. Program codes with sample output of all assignments are to be submitted in softcopy.

**Guidelines for Laboratory/Term Work Assessment:**

1. Continuous assessment of laboratory work will be based on overall performance of Laboratory assignments by a student.
2. Each Laboratory assignment will be assessed based on the rubric approved by DUPC.

**Guidelines for Laboratory Conduction:**

1. Set of suggested assignments is provided for reference. Lab instructors may design the assignments with variations or suitable updates as required.
2. Operating System recommended: 64 -bit Open-source Linux or its derivative.
3. Programming tools recommended: Open-source/C/C++/JAVA/Python

**Detailed Syllabus**

| Unit | Description | Duration (Hrs) |
|---|---|---|
| I | **Introduction to Operating Systems**: Introduction, Evolution, Types, Services, system calls.<br>**Process Management**: Process Concept -Process states, Process control block, Threads -Introduction, POSIX / pthreads. Process Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms. | 7 |
| II | **IPC and Process Synchronization**: Inter process Communication mechanisms: Pipes, Shared memory, Message passing Process Synchronization: Introduction, Critical-Section Problem, Hardware Support for Synchronization, Mutex Locks, Semaphores, Synchronization problem: Reader-writer, producer-consumer problem, Dining Philosophers problem.<br>**Deadlocks:**<br>Introduction, Methods for Handling Deadlocks. | 8 |
| III | **Memory management:**<br>Introduction, Contiguous and non-contiguous, Swapping, Memory Allocation Strategies, Paging, Segmentation, Virtual Memory: Background, Demand paging, Page Replacement Policies. | 8 |
| IV | **I/O Management**<br>Concept of Files, File Allocation Methods, Free -Space Management. Disk Scheduling- FIFO, SSTF, SCAN, C-SCAN. | 7 |
| | **Self Study**<br>Study the corresponding components and services of contemporary Linux distribution, research paper reading. | 6* |
| | **Total** (*Not Included) | **30** |

**Suggested Lab Assignments:**

1. A set of processes is to be executed on a processor. Implement scheduling FCFS, SJF, Round Robin and Priority algorithms to identify the optimal schedule for executing these processes.
2. Various in-build inter-process communication mechanisms (like POSIX, FIFO, FUTEX, SysV) are available in the Linux based operating systems. Simulate such inter-process communication mechanisms using pipes.
3. Write a program to demonstrate the reader-writer/ producer- consumer/ dining philosophers synchronization problem and its solution.
4. Write a program to compare paging replacement algorithms:
   a) FCFS
   b) Least Recently Used (LRU)
   c) Optimal algorithm
5. Write a program to compare disk scheduling algorithms: FIFO, SSTF, SCAN, C - SCAN

**Text Books:**

1. 1. Silberschatz, Galvin, Gagne, "Operating System Principles", 10th Edition,2018, Wiley, ISBN 978 – 1 – 118 - 063330
2. Stallings W., "Operating Systems - internals and design principles", 9th Edition - 2018, Pearson, ISBN - 13: 978 – 013 – 467 – 0959
3. H.M. Deitel, P. J. Deitel, D. R. Choffnes, “Operating Systems “, Pearson, 3rd Edition, ISBN 0131828274, 97801318282